Adaptive Multilooking of Multitemporal Differential SAR Interferometric Data Stack Using Directional Statistics

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***Abstract*— In this article, we present an innovative space–time adaptive multilooking technique that operates on a sequence of multitemporal, differential synthetic aperture radar interferograms. The developed approach relies on the application of the fundamentals of directional statistics theory. At variance with other methods that identify the set of statistically homogenous pixels (SHPs) within a multilooking (complex averaging) window based on the statistics of the single-look-complex (SLC) SAR images, the proposed method is exclusively based on the analysis of the multitemporal sequence of full resolution DInSAR interferograms. The SHPs are then used to generate spatially adaptive multilooked interferograms both at the native, full-scale grid of the SLC images and at the multilooked resolution scale. The algorithm is effective and simple to implement, only requiring the availability of a sequence of full-scale differential SAR interferometry (DInSAR) interferograms. The interferograms can then be used to generate ground displacement time-series through advanced multitemporal interferometric SAR (MTInSAR) approaches. Experimental results obtained by applying the adopted technique to two SAR data sets acquired at X- and L-band, respectively, demonstrate the validity of the developed method.**

***Index Terms*— Circular statistics, differential synthetic aperture radar (SAR) interferometry (DInSAR), distributed scatterers (DSs), surface deformation, SAR.**

I. Introduction

VARIOUS Earth's surface deformation phenomena connected to both natural [1]–[13] and human-induced processes [14]–[17] have successfully been detected and extensively studied in the last two decades using the differential synthetic aperture radar interferometry (DInSAR) technique [18], [19], which relies on the measurement of the phase difference (i.e., interferogram) between two SAR acquisitions made by an SAR instrument at different times. DInSAR provides a measurement of the ground displacement projected onto the radar sensor line-of-sight (LOS) direction with centimeter-to-millimeter accuracy [20]. Although the DInSAR technique was initially applied to investigate single deformation episodes [19], [21]–[23], a major advancement was made by the development of multitemporal interferometric SAR (MTInSAR) techniques [24]–[32] that allow the detection and continual monitoring of the temporal evolution of surface displacements in areas not significantly affected by temporal decorrelation [33]. These algorithms are based on appropriate combination of the sequences of DInSAR interferograms generated from multiple SAR acquisitions.

The MTInSAR techniques are generally grouped into two main categories of the persistent scatterers (PSs) [24]–[26] and small baseline (SB) methods [27]–[31], although one hybrid solution that incorporates both PS and SB strategies has been implemented [32]. The PS techniques operate on the pixels of single-look complex (SLC) SAR images that are representative of resolution cells characterized by a dominant scatterer and hence, with high phase stability in the entire time series. Conversely, the SB methodologies extract information from distributed scatterers (DSs) on the ground that may be affected by spatial and temporal decorrelation [33]. For SB methods, the multiple-master InSAR data pairs that have small values of the temporal and spatial baseline preserve coherence and are suitable to extract useful information from the DSs. Additionally, the SB interferograms are usually multilooked (complex averaged) [34], [35] to further reduce the decorrelation noise. Among the SB techniques, a well-known approach is the Small Baseline Subset (SBAS) [27], which was originally developed to work with multilooked data and subsequently extended to work with single-look interferograms [28] with the aim of also detecting accurate deformation signals for single buildings or man-made structures. An improved SBAS processing chain that complements a proper SB interferogram selection and E-MTInSAR noise-filtering technique with the extended minimum cost flow (EMCF) space–time phase unwrapping methodology [36]–[38] has also been proposed [39].

Recently, the SqueeSAR technique [40] and other alternative multitemporal approaches [41]–[44] have been proposed for

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the characterization of DS targets. The SqueeSAR method [40] operates on all possible differential SAR interferograms computed at the same spacing as the single-look ones, but following a low-pass (LP) phase filtering step. The method is based on the preliminary selection of statistically homogenous pixels (SHPs) that identify the location of dominant DS objects within every multilooking resolution cell, over which the adaptive multilooking operations are subsequently carried out. The selection of the SHPs is based on application of the two-sample Kolmogorov–Smirnov (KS) test [45] to SAR amplitude data vectors for pairs of neighboring SAR pixels. A maximum likelihood (ML) estimator [46] is applied to the multilooked interferograms generated by averaging over the identified SHPs.

In this article, we present a new approach for the adaptive spatial multilooking [47]–[49] of a stack of time-redundant, multitemporal DInSAR interferograms. In contrast to SqueeSAR, the proposed technique does not use the amplitude signal as a proxy for the phase stability. Rather, it directly investigates the phase stability of a sequence of differential SAR interferograms to identify statistically homogenous pixels by invoking the fundamentals of directional statistics [50]. The generated multilooked phase vectors are characterized by reduced phase dispersion and therefore are less corrupted by decorrelation noise. The stack of noise-filtered multilooked DInSAR interferograms can be used to compute surface displacement time-series using any MTInSAR method working with multilooked interferograms [6], [38], [39].

This article describes the theoretical basis for the new adaptive multilooking method based on the interferometric phase, discusses its expected performance improvement, and presents a detailed description of its implementation. The effectiveness of the method is demonstrated through two experiments carried out on sequences of SAR images collected at C- and L-band. The article is organized as follows. Section II describes the process applied for the identification of the SHPs. Sections III and IV discuss the performance of the adaptive multilooking approach and the generation of the multilooked interferograms, respectively. Experimental results are presented in Section V and conclusions and future perspectives are provided in Section VI.

II. Identification of SHPs

The developed approach for identifying the SHPs, described in this section and briefly presented first in [51], identifies the SHPs by searching for those pixels for which the vectors of the wrapped phases are less dispersed with respect to the estimated (averaged) multilooked phases within a multilooking window. The statistical homogeneity of the identified group of pixels is tested by applying Kuiper's statistical test [52] to the directional data vectors of their wrapped phases. Once the SHPs are identified in every multilooking window, the vector of the multilooked phase is obtained by adaptively averaging the single-look DInSAR interferograms over the SHP pixels.

Specifically, let us consider a set of N SAR images acquired over the same area on Earth at different epochs $[t_0, t_1, \ldots, t_{N-1}]^T$ properly coregistered to a master reference image. From them, a sequence of M single-look differential SAR interferograms, namely $\phi(P) \equiv [\phi_0(P), \phi_1(P), \ldots, \phi_{M-1}(P)]^T$, is generated, with $P \equiv (x, y)$ being the azimuth and range coordinates of a generic radar pixel of the analyzed scene. The proposed adaptive multilooking technique is based on the key observation that all SAR pixels in an averaging (multilooking) spatial window are not necessarily statistically homogeneous because they may represent a mix of different distributed targets on the ground. Accordingly, if the spatial multilooking operations are performed independently (i.e., interferogram by interferogram) by means of a boxcar window and by averaging together all the phase signals of pixels within that window, the resulting low-pass (LP) wrapped phase vector (i.e., the vector of the wrapped multilooked phases) will in general not be well estimated. In order to increase the signal-to-noise ratio (SNR), only phase data from SHPs (over which the unknown multilooked phases as well the coherence values can be assumed constant) should be averaged.

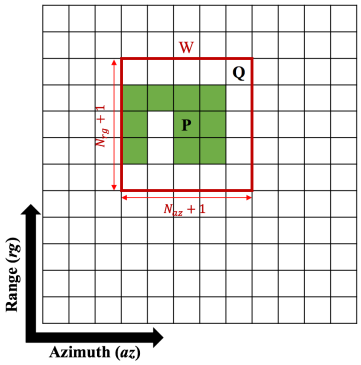


Fig. 1. Averaging boxcar window W centered on the location of SAR pixel $P(x, y)$. The green area shows the location of the SHPs related to pixel P, identified through the method described in Section II.

Unlike the methods proposed in [40], [41], and [49], which require statistical analysis of the amplitude values of the SAR images used in interferogram generation, our method exclusively makes use of sources of information retrievable from the interferograms themselves, namely directional data represented by the sequence of differential interferometric phases in the spatial averaging (multilook) window. To create the right framework for the presentation of the algorithm, let us describe the proposed method by referring to a generic radar pixel of coordinates $P \equiv (x, y)$ in the averaging spatial window, W, centered on the location of P (see Fig. 1) and with dimensions $N_{\text{az}} + 1$ and $N_{\text{rg}} + 1$ in the azimuth and range directions, respectively. The phase values computed at the $(N_{\text{az}}+1) \times (N_{\text{rg}}+1)$ SAR pixels locations in the averaging window W and related to the kth interferogram are represented by the following complex directional data vector:

$$\{\exp[j\phi_k(Q)]\}_{Q \in W, k=0,1,\ldots,M-1} \tag{1}$$

where $j = \sqrt{-1}$. From [50], we know that the mean direction ϕ_k^{LP} of the kth DInSAR interferogram represents the direction

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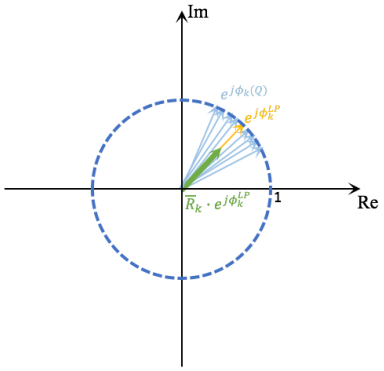


Fig. 2. Representation of the directional data phase distribution on a unit circle in the complex plane.

of the mean resultant vector

$$\prod_k = \frac{1}{(N_{\text{az}}+1)\times(N_{\text{rg}}+1)}\sum_{Q\in W}\exp\left[j\phi_k(Q)\right] = \bar{R}_k\cdot\exp\left[j\phi_k^{\text{LP}}\right] \tag{2}$$

where $\bar{R}_k$ is the mean resultant length of the kth directional data phase distribution (see Fig. 2).

Note also that

$$\cos[\phi_k^{\text{LP}}] = \frac{\bar{C}_k}{\bar{R}_k};\quad \sin[\phi_k^{\text{LP}}] = \frac{\bar{S}_k}{\bar{R}_k} \tag{3}$$

where

$$\bar{C}_k = \frac{1}{(N_{\text{az}}+1)\times(N_{\text{rg}}+1)}\sum_{Q\in W}\cos\left[\phi_k(Q)\right] \tag{4a}$$

$$\bar{S}_k = \frac{1}{(N_{\text{az}}+1)\times(N_{\text{rg}}+1)}\sum_{Q\in W}\sin\left[\phi_k(Q)\right] \tag{4b}$$

$$\bar{R}_k = \sqrt{\bar{C}_k^2+\bar{S}_k^2} \tag{4c}$$

$$\phi_k^{\text{LP}} = \text{arctg}\left[\frac{\bar{S}_k}{\bar{C}_k}\right]. \tag{4d}$$

It can be shown [50] that the circular variance ν_k of the kth directional data is defined as

$$\nu_k = 1-\left|\prod_k\right| = 1-\bar{R}_k \tag{5}$$

and is equal to the minimal value of the phase dispersion of the directional data with respect to its mean direction, ϕ_k^{LP}. Accordingly, if $\nu_k = 0$ then the directional data is concentrated on ϕ_k^{LP}. Conversely, when $\nu_k \to 1$ the distribution is so scattered that there is no concentration around any direction, and the selected patch W is fully corrupted by noise [33].

We note that the mean resultant length of the kth DInSAR phase data stack, namely, $\bar{R}_k = 1-\nu_k$ approaches the spatial coherence value of the corresponding kth interferogram [33], [53]

$$\gamma_k = \left|\frac{E\left[i_{M,k}\cdot i_{S,k}^*\right]}{\sqrt{E\left[\left|i_{M,k}\right|^2\right]\cdot E\left[\left|i_{S,k}^*\right|^2\right]}}\right| \tag{6}$$

where $i_{M,k}$ and $i_{S,k}$ are the master and slave complex-valued SAR images form the kth interferogram, respectively, and $E[\cdot]$ represents the statistical expectation operator [54]. Such an outcome was originally addressed in [55], and it is discussed in more depth in Appendix A. More specifically, Lopez-Martinez and Fabregas [55] demonstrated that $\bar{R}_k = 1-\nu_k$ is monotonically increasing with respect to γ_k. Accordingly, the mean resultant length of the directional data $\{\exp[j\phi_k(Q)]\}_{Q\in W,k=0,1,\ldots,M-1}$ is an alternative interferometric phase parameter for the spatial coherence. The advantage in using it is that it does not depend on the amplitude values of the master and slave images forming the interferogram. Accordingly, it is well suited to the analyses proposed in this article, which exploits observations that are unit vectors in the complex plane.

Below, we present a procedure to identify within every multilooking window the set of SAR pixels that are statistically homogeneous and are representative of a DS. In our method, the SHPs are identified by directly exploiting phase information using directional statistics fundamentals [50], [56], [57].

SHP identification is accomplished through a two-step processing procedure. We preliminarily observe that the phase signature associated with a DS target within the given averaging (multilooked) window is a random process whose different realizations are represented by the phase values measured for every DS target pixel in $\Omega \subseteq W$. Every DS has a specific (unknown) multilooked phase and relevant statistical properties. As a starting point, let us consider the averaging window W centered on the location of the SAR pixel with coordinates $P \equiv (x, y)$ (Fig. 1). If all SAR pixels of W are averaged together, we can obtain, for every interferogram, an estimate of the multilooked phase. From (4d), the estimated LP, multilooked phase data vector is

$$\exp\left[j\phi^{\text{LP}}\right] \equiv \left\{\exp\left[j\phi_0^{\text{LP}}\right],\exp\left[j\phi_1^{\text{LP}}\right],\ldots,\exp\left[j\phi_{M-1}^{\text{LP}}\right]\right\}. \tag{7}$$

The key element of the developed adaptive multilook operation is to compare the statistical phase dispersion of the (directional) data representing the single-look DInSAR phase vectors related to every pixel in the averaging window W to the vector of LP phase components defined in (7).

If all SAR pixels in W are characterized by phase data $\{\exp[j\phi_k(Q)]\}_{Q\in W,k=0,1,\ldots,M-1}$ that are less dispersed with respect to $\exp[j\Phi^{\text{LP}}]$, then all SAR pixels can be inferred to belong to the same DS on the ground. Conversely, if some SAR pixels have large phase dispersion with respect to $\exp[j\Phi^{\text{LP}}]$, statistically they do not belong to the same homogenous DS. These pixels can be rejected from W, and the location of the SHP, namely Ω (see Fig. 1) is identified by the remaining pixels.

Step 1: First, we compute the stack of the high-pass phase signals, for every interferogram, by multiplying $\{\exp[j\Phi(Q)]\}_{Q\in W}$ by the complex conjugate of $\exp[j\Phi^{\text{LP}}]$

$$\begin{aligned}&\exp[j\Phi^{\text{HP}}]_{Q\in W,k=0,1,\ldots,M-1}\\ &\equiv\left\{\exp[j\phi_k(Q)]\cdot\exp\left[-j\phi_k^{\text{LP}}\right]\right\}_{Q\in W,k=0,1,\ldots,M-1}\\ &\equiv\left\{\exp\left[j\phi_k^{\text{HP}}\right]\right\}_{Q\in W,k=0,1,\ldots,M-1}.\end{aligned} \tag{8}$$

At this stage, we can simply demonstrate that the high-pass phase components of every interferogram are concentrated

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around zero. In particular, if we consider a generic pixel, namely Q, that belongs to the averaging multilook window, the mean resultant vector of the directional data $\exp[j\phi_k^{\mathrm{HP}}(Q)]$ is obtained as follows:

$$\prod(Q) = \frac{1}{M}\sum_{k=0}^{M-1}\exp\left[j\phi_k^{\mathrm{HP}}(Q)\right] = \bar{R}(Q)\cdot\exp[j\mu(Q)]. \quad (9)$$

From [50], it is easy to show that the circular variance of the phase distribution

$$v_k = 1 - |\Pi(Q)| = 1 - \bar{R}(Q) \quad (10)$$

represents the minimal phase dispersion of the phase distribution $\exp[j\phi_k^{\mathrm{HP}}(Q)]$ with respect to its mean direction $\mu(Q)$. In particular, it can be proven that $\mu(Q) = 0$. Accordingly, if the circular variance $v(Q)$ is small, then the high-pass phase components related to the SAR pixel Q have a very small phase deviation with respect to zero, thus implying that the phase $\{\exp[j\phi_k(Q)]\}_{Q\in W,k=0,1,\ldots,M-1}$ is less dispersed w.r.t. the estimated multilooked phase data Φ^{LP}.

Let us now give some additional information on the statistics of the high-pass data vector. Under the hypothesis that the SAR phase data have a zero-mean multivariate circular normal distribution [46], the probability density function (pdf) of the n-multilooked $\phi^{(n)}$ interferometric phases, as independently derived from several authors [53], [58]–[61], is known to be

$$p_{\phi^{(n)}}(\phi^{(n)}) = \frac{\Gamma(n+\frac{1}{2})(1-\gamma^2)^n\gamma\cos(\phi^{(n)}-\bar{\phi}^{(n)})}{2\sqrt{\pi}\Gamma(n)(1-\gamma^2\cos^2(\phi^{(n)}-\bar{\phi}^{(n)}))} + \frac{(1-\gamma^2)^{(n)}}{2\pi}\cdot F\left(n,1;\frac{1}{2};\cos^2(\phi^{(n)}-\bar{\phi}^{(n)})\right) \quad (11)$$

where γ is the amplitude of the complex correlation coefficient (i.e., the coherence) of the interfering SAR images [see also (6)]; $\bar{\phi}$ and $\bar{\phi}^{(n)}$ are the expected (noise-free) values of the single- and multilook phases, respectively; n is the number of independent looks, and $\Gamma(\cdot)$ and $F(\cdot)$ are the gamma and the Gauss hypergeometric functions, respectively.

To statistically characterize the high-pass phase data vectors, we here invoke the characteristic functions of distributions on the circle. Under the simplifying hypothesis that high-pass phase signals are time-independent, it can be assumed that the high-pass phase is distributed with a von Mises pdf [62]. For instance, if we consider the directional data $[\theta_1,\theta_2,\ldots,\theta_M]$ with mean resultant length $\bar{R}$ and mean direction μ calculated as

$$\frac{1}{M}\sum_{k=0}^{M-1}\exp[j\theta_k] = \bar{R}\exp[ju] \quad (12)$$

then the von Mises distribution of that directional data has the following pdf:

$$p_{\mathrm{VM}}(\theta;\mu,k) = \frac{1}{2\pi I_0(\kappa)}\exp(\kappa\cos(\theta-\mu)) \quad (13)$$

where $I_0(\cdot)$ is the modified Bessel function of the first kind and order zero [63] and κ is the concentration parameter of the von Mises distribution. Note that for very small values of the concentration parameter κ the von Mises distribution approximates a uniform distribution, whereas for larger values of the concentration parameter κ, it tends to become more concentrated at the point $\theta = \mu$, as shown in Fig. 3.

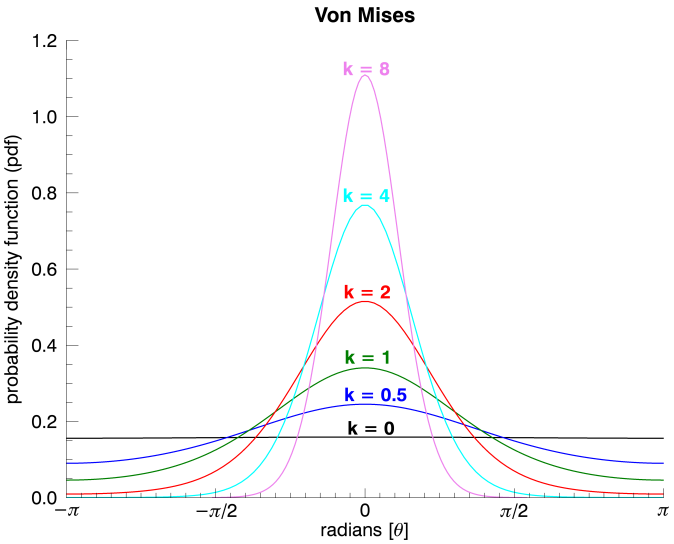


Fig. 3. Von Mises's pdf for different values of κ.

Assuming that the high-pass phase vector related to the given SAR pixel Q is von Mises distributed with concentration parameter κ, it can be shown [50] that the first moment of $\bar{R}$ is

$$E[\bar{R}(Q)] = \frac{I_1(\kappa)}{I_2(\kappa)} = A(\kappa) \cong 1 - \frac{1}{2\kappa} - \frac{1}{8\kappa^2} \quad (14)$$

where $I_1(\cdot)$ is the modified Bessel function of the first kind and first order. From (14), it is evident that the measured value of the mean resultant length of the directional data [see (12)] provides an estimate of the (unknown) concentration parameter κ of the Von Mises distribution and statistically characterizes the phase. In particular, the distribution of the high-pass phase signals is more or less concentrated about its mean direction μ, which is equal to zero by design. This means that large values of the measured $\bar{R}$ correspond to SAR pixels with phases more concentrated around the estimated LP (multilook) phase vector $\exp[j\Phi^{\mathrm{LP}}]$. A suitable approximation of the Von Mises distribution for large values of the concentration parameter κ is the wrapped normal distribution, $WN(\mu,\rho)$ with $\rho = \bar{R}$ obtained by wrapping onto the circle the normal distribution, $N(\mu,\sigma^2)$, with

$$\rho = \exp\left[-\frac{\sigma^2}{2}\right]. \quad (15)$$

Accordingly, there is a direct relationship between the measured value of $\rho = \bar{R}$ and the standard deviation of high-pass phase signal. From (15), we can derive the mathematical relationship between the measured circular variance of the random high-pass phase distribution and the standard deviation of the wrapped-normal distribution, which is $\sigma^2 = -2\ln\rho = -2\ln(1-v)$, where v is the circular variance related to the given analyzed pixel Q in the averaging window (see Fig. 1).

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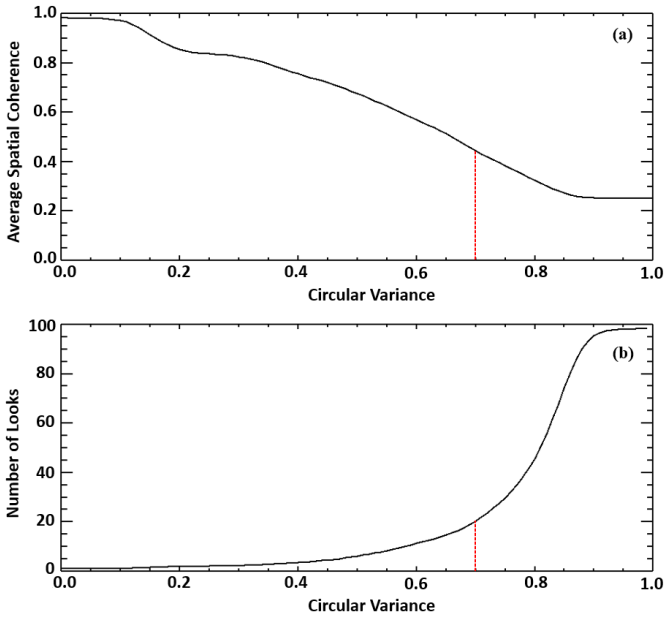


Fig. 4. Plot of (a) average spatial coherence and (b) number of looks versus the applied circular variance threshold.

Step 2: Second, we identify the pixels that are representative of a DS on the ground by means of two operations: The first relies on the assumption that the DS target location is defined as the locus of pixels for which the circular variance does not exceed a fixed threshold, namely $\Omega = \{Q : \nu(Q) < \alpha\}$ [51] (from experimental tests and theory considerations addressed in Section III, a value of $\alpha = 0.7$ was found suitable). The second operation is to check the statistical homogeneity of the selected pixels. This is done by selecting that pixel exhibiting the lowest circular variance value, namely Q_{REF}, within the group of SAR pixels Ω satisfying the threshold on the circular variance, and then applying the two-sample Kuiper's test [50], [52], [64], [65] to the pairs of pixels $Q - Q_{\text{REF}}$ where $Q \in \Omega$. We note that the two-sample Kuiper's test is a goodness-of-fit test [52]. Kuiper's test is a modification for circular data of the KS test [45], [50], [54]. The goodness of the Kuiper's test has also been assessed [64]. Specifically, the used implementation of Kuiper's test returns the probability associated with the Kuiper's statistic in terms of the p-value [64], with small values indicating that $S_{Q_{\text{REF}}}$ and S_Q are significantly different. In our case, we used a threshold of p-value equal to 0.20. Finally, only those candidate SAR pixels in Ω that are not rejected by the Kuiper's test are considered as SHP and used for estimating the multilooked phases, interferogram by interferogram, as follows:

$$\phi_k^{\text{ML}} = \arctan\left(\frac{\sum_{Q\in\text{SHP}} \sin[\phi_k(Q)]}{\sum_{Q\in\text{SHP}} \cos[\phi_k(Q)]}\right) \quad \forall k = 0, \ldots, M-1. \tag{16}$$

For every interferogram, we consider the following parameter, hereinafter referred to as circular coherence, as an interferometric phase quality indicator for the computed adaptive

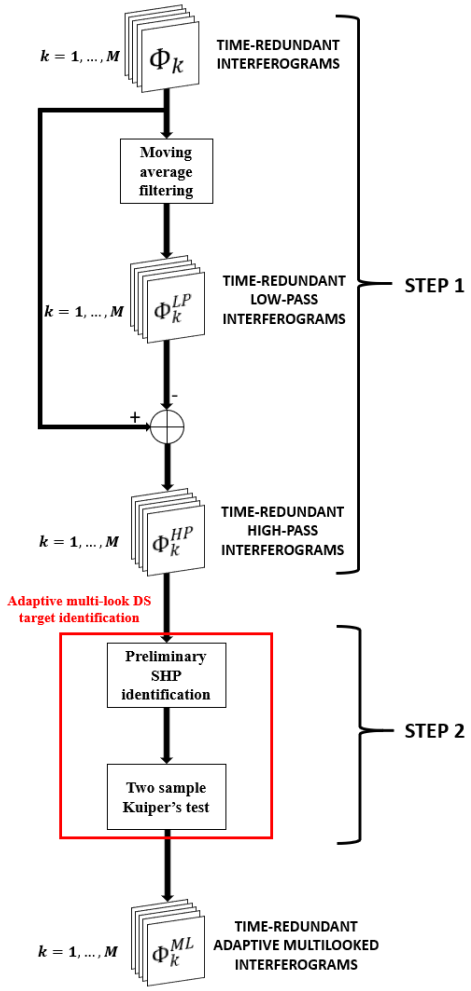


Fig. 5. Block diagram of the proposed adaptive multilooking DS identification technique. We have highlighted the steps 1 (8) and 2 [(18) and (17)] of the algorithm, which are fully detailed in Section II.

multilooked phase at the SHP's location

$$\gamma_k^{\text{ML}} = \frac{1}{|\Omega'|}\left|\sum_{Q\in\Omega'} \exp\left[j\phi_k(Q)\right]\right| \tag{17}$$

where Ω' represents the (actual) group of SHPs that have passed the Kuiper's test, and $|\Omega'|$ is its cardinality.

As stated previously, the circular coherence approximates the spatial coherence of the kth interferogram (see Appendix A) and gives an indication of the amount of noise that corrupts the computed multilooked phases at the SHP' locations.

III. SHP Identification Algorithm Performance

In this section, we analyze the computed stack of multitemporal multilooked phases at SHPs in terms of the expected improvement of the average circular coherence of the phase data stack. As shown in Section II (step 2) the strategy adopted to identify the SHPs in a given averaging window is based on identifying and discarding those pixels with a circular variance that is larger than a selected threshold, α. Hence, for every averaging multilooking window W, the group of pixels $\Omega = \{Q : \nu(Q) < \alpha\}$ is identified. Kuiper's test is

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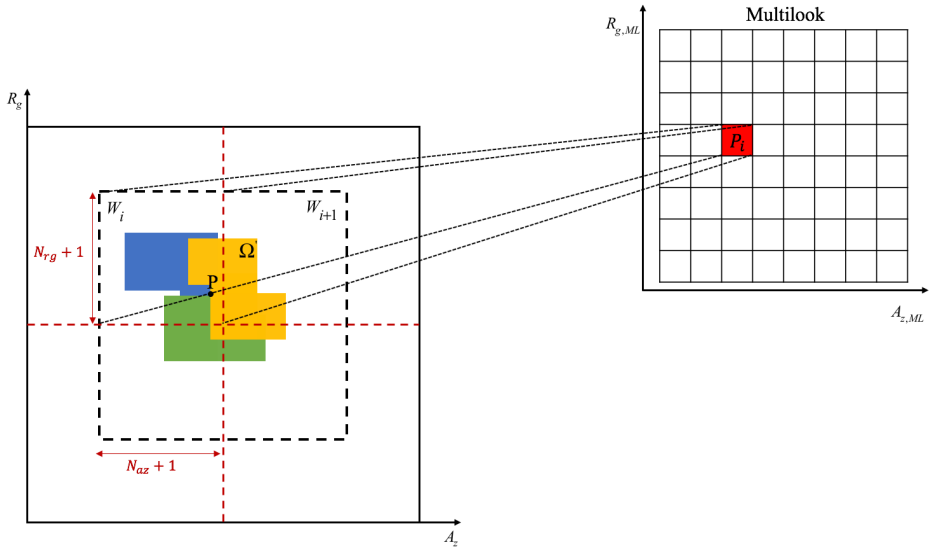


Fig. 6. Interferogram spatial grid decomposition in a set of nonoverlapped multilooked rectangular patches.

subsequently applied to Ω to discard other pixels that are likely not characterized by the same statistical distribution, thereafter obtaining the group $\Omega' \subseteq \Omega \subseteq W$. First, we note that over Ω' the computed average spatial coherence of the used interferograms, $\bar{\Gamma}$, increases after the application of the procedure outlined in Section II relative to the value computed by considering all pixels in the averaging box W. In particular, it can be proven (see Appendix B) that

$$\bar{\Gamma}_{\Omega'}(Q) \geq \bar{\Gamma}_{W}(Q) \quad \forall Q \in \Omega' \tag{18}$$

where

$$\bar{\Gamma}_{\Omega'}(Q) = \frac{1}{M}\sum_{k=0}^{M-1}\frac{1}{|\Omega'|}\left|\sum_{Q\in\Omega'}\exp[j\phi_k(Q)]\right| \tag{19a}$$

$$\bar{\Gamma}_{W}(Q) = \frac{1}{M}\sum_{k=0}^{M-1}\frac{1}{|W|}\left|\sum_{Q\in W}\exp[j\phi_k(Q)]\right| \tag{19b}$$

are the (average) circular coherence of the entire multitemporal stack of M DInSAR multilook interferograms as calculated by averaging the phase values on Ω' (19a) or W(19b), and corresponds to applying the proposed adaptive multilooking approach or a conventional boxcar window filtering, respectively. Equation (18) states that the phase stability of the adaptive multilook phase is enhanced with respect to the conventional case, and as a consequence the decorrelation noise artifacts are reduced [33].

To confirm the validity of (18) with real SAR data, we have performed an experiment using the stack of DInSAR interferograms shown in Section V for the Mt. Etna case study. More specifically, we have extracted a patch from the investigated zone (see Fig. 8) and computed the upper bound values of the (average) circular coherence for the entire stack of data as a function of the adopted threshold for the identification of the SHPs (see Appendix B for its computation).

For this computation, the circular coherence values of all pixels of the selected patch have been considered. Fig. 4(a) shows the plot of the (average) spatial coherence versus the selected threshold for the considered patch, and Fig. 4(b) shows the plot of the (average) number of looks versus the adopted threshold.

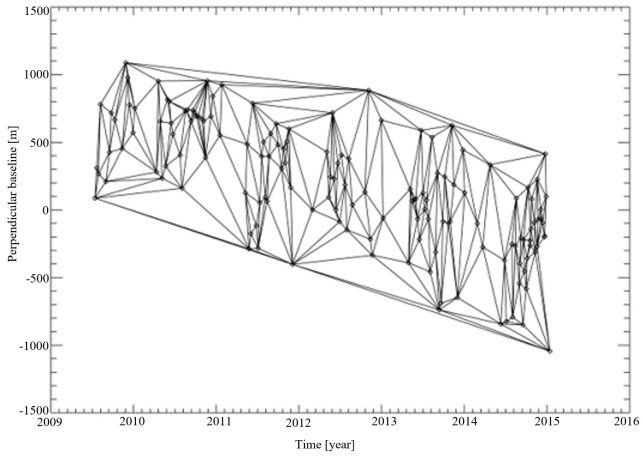


Fig. 7. Distribution of the available SAR data used for the Mt. Etna case study in the temporal/perpendicular baseline plane. Diamonds indicate the SAR acquisitions, and edges indicate the interferometric SAR data pairs, respectively.

Note that as the threshold α decreases, the number of independent samples that are averaged together drastically decreases [see Fig. 4(b)], thus making the coherence value biased [53]. Therefore, a tradeoff exists between the circular coherence increase and the number of averaged samples. A strategy that may be adopted to identify the correct threshold to be used is, for instance, to identify the elbow point of the curve shown in Fig. 4(b) and select a threshold exceeding that point. In our case, the elbow point corresponds to a threshold of the circular variance that is equal to 0.7.

Potential users can select different thresholds, but taking into account the mentioned tradeoff. Considering the plots in Fig. 4(a) and (b) for the experiments shown in Section V we have selected 0.7 as a suitable threshold. Further analyses would be required to fix the optimal threshold, and this is a matter for future investigations.

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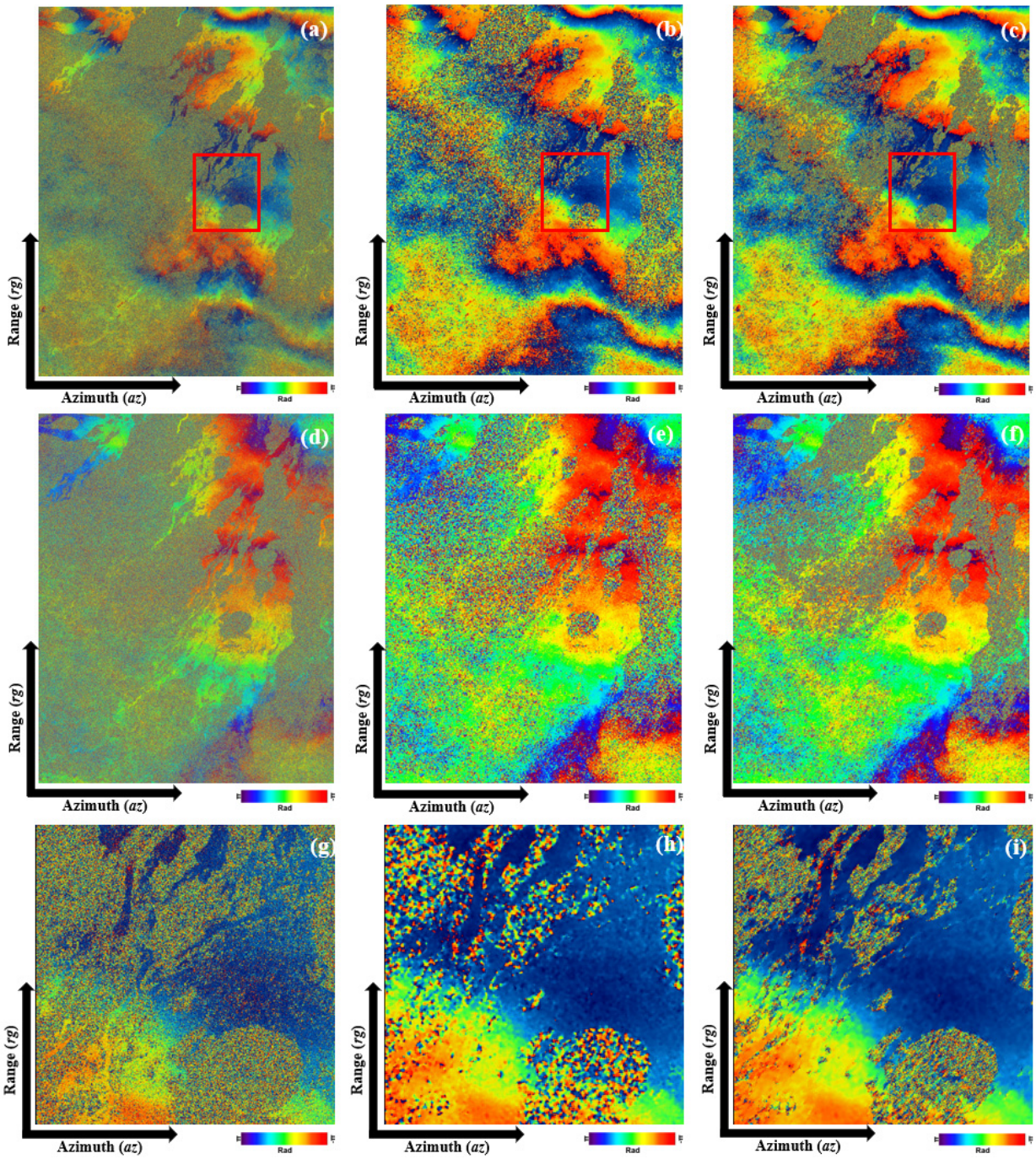


Fig. 8. Experimental results of Mt. Etna volcano. Unfiltered CSK full resolution interferograms of (a) 09/05/2012–10/06/2012 and (d) 27/06/2012–26/06/2012 interferometric SAR data pairs. (b) and (e) Noise-filtered interferograms related to (a) and (d), respectively, obtained using a conventional boxcar window. (c) and (f) Adaptive noise-filtered interferograms corresponding to (a) and (d), respectively. (g)–(i) Zoomed views of (a)–(c) at the area identified by the red rectangular in (a)–(c), respectively.

IV. Generation of the Adaptive Multilook Interferograms

In this section, we outline the overall procedure to generate spatially adaptive multilook interferograms. As stated in Section II, for every interferogram, the computed value of the multilooked phase in the selected averaging window is, first, associated with every pixel forming the identified SHPs for every averaging window (see Fig. 1). However, the information on the multilooked phase of a generic pixel, namely P, is retrieved by the strategy outlined in Section II. To this end, the process described in steps 2 and 3 is, then, repeated for all pixels $\{P \in A_z \times R_g\}$ of the azimuth/range spatial grid by progressively selecting the relevant averaging (moving) windows centered on the given SAR pixel, identifying the SHPs in the corresponding averaging windows, and eventually obtaining an estimate of the multilooked phases at the location of SHPs, as provided by (16).

After scanning the entire $A_z \times R_g$ grid of the analyzed scene, for every pixel P, we will have Θ different groups of SHPs to which the pixel P belongs. The vector of the adaptive

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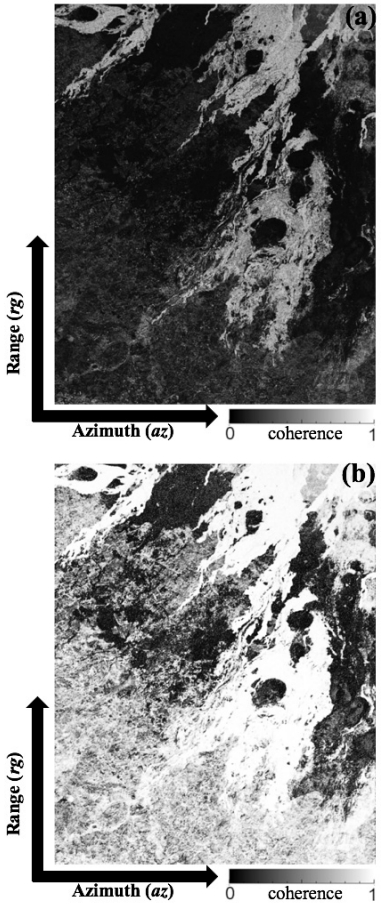


Fig. 9. (a) Average spatial coherence for the original and (b) adaptively multilooked Mt. Etna interferograms.

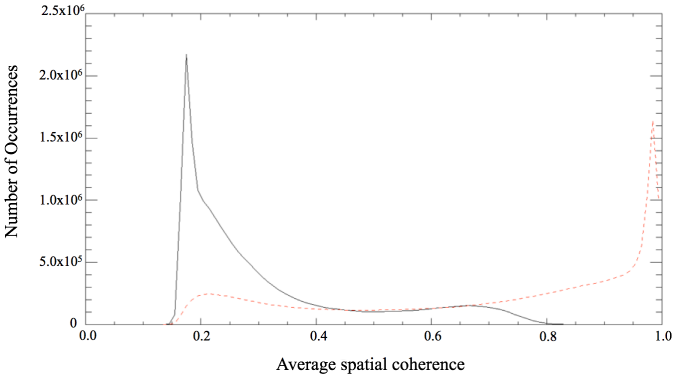


Fig. 10. Histogram of spatial coherence of original (black continuous line) and adaptively multilooked (red dashed line) interferograms.

multilooked phases is then computed for every pixel P as follows:

$$\phi_k^{\mathrm{ML}}(P) = \arctan\left(\frac{\sum_{h=1}^{\Theta}\sum_{Q\in\Omega'}\sin[\phi_k(Q)]}{\sum_{h=1}^{\Theta}\sum_{Q\in\Omega'}\cos[\phi_k(Q)]}\right)$$
$$P \in \bigcap_{h=1}^{\Theta}\Omega'_h, \quad k=0,1,\ldots,M-1. \tag{20}$$

The whole process is pictorially represented in Fig. 5. A few additional considerations are now in order. First, we observe that for the computation of the multilooked phase values

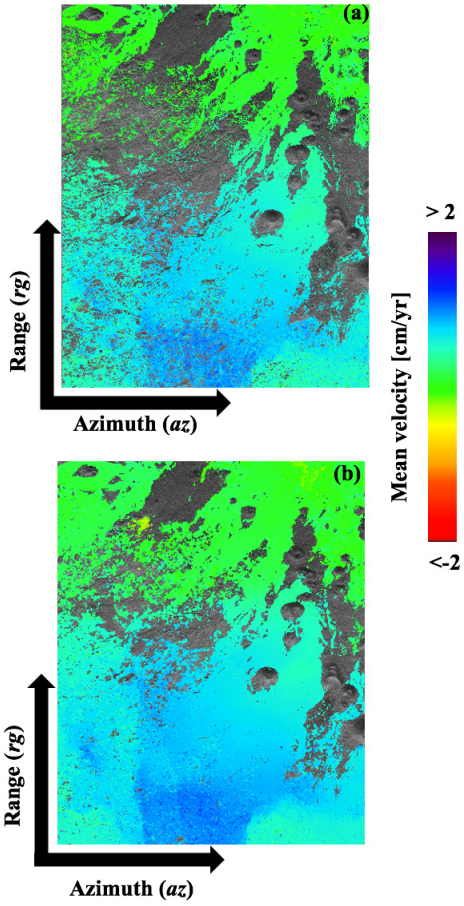


Fig. 11. Mean deformation velocity map of Mt. Etna area (a) before applying the adaptive multilook algorithm and (b) after.

of every analyzed SAR pixel, Θ different SHPs are identified and exploited. Accordingly, the number of independent looks, which is adaptively chosen, varies from pixel to pixel. Of course, most of the averaged phases relevant to the selected SAR pixels are dependent, thus the number of effective looks is significantly smaller than $\Theta\times(N_{\mathrm{az}}+1)\times(N_{\mathrm{rg}}+1)$. Second, although the phases in (20) related to every SAR pixel are multilooked, they are eventually represented on the native grid of the single-look SLC SAR images. In most operational use, however, it is cumbersome to work with full-scale multilooked interferograms computed at the grid of SLC SAR images.

Therefore, in some cases it is more useful to have multilooked interferograms directly generated at the degraded multilooked spatial grid. In this case, the whole spatial grid $A_z\times R_g$ is decomposed into a set of nonoverlapped multilooked rectangular patches, namely W_i, with size $(N_{\mathrm{az}}+1)\times(N_{\mathrm{rg}}+1)$ (see Fig. 6). For every rectangular patch W_i, corresponding to the radar pixel of multilooked coordinates $P_i(A_{z,\mathrm{ML}}\times R_{g,\mathrm{ML}})$ the multilooked phase estimates are

$$\tilde{\phi}_k^{\mathrm{ML}}(P_i) = \arctan\left(\frac{\sum_{h=1}^{\Theta}\sum_{Q\in\Omega'_k}\sin[\phi_k(Q)]}{\sum_{h=1}^{\Theta}\sum_{Q\in\Omega'_k}\cos[\phi_k(Q)]}\right)$$
$$P_i\cap\Omega'_h\neq\emptyset, \quad h=1,2,\ldots,\Theta, \quad k=0,1,\ldots,M-1. \tag{21}$$

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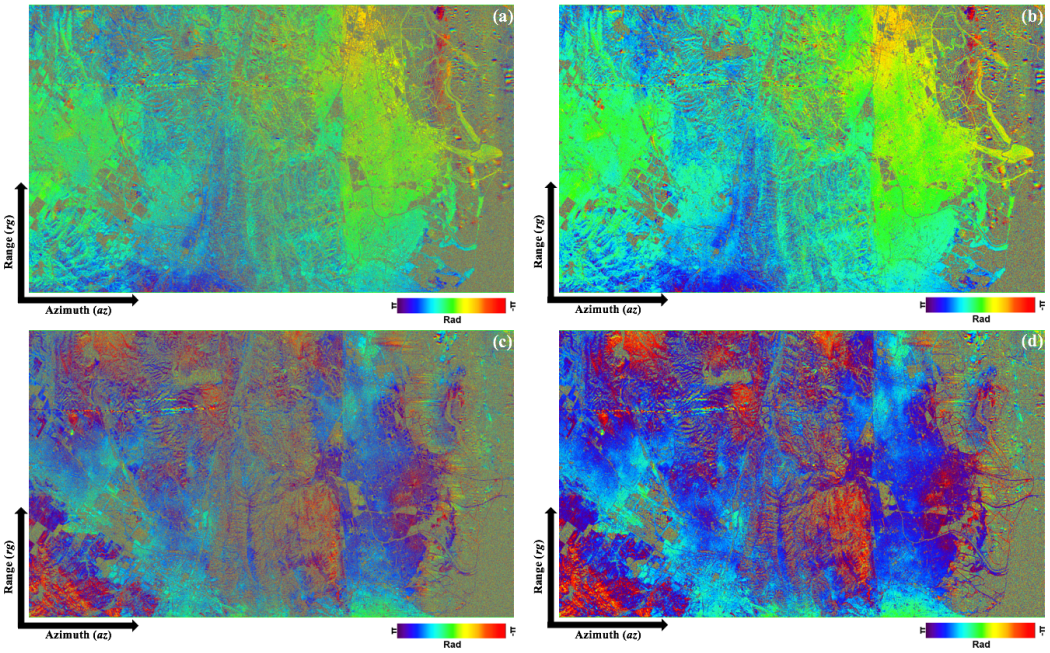


Fig. 12. UAVSAR InSAR multilooked interferogram related to the Haywaard fault case study. (a) and (c) Left and (b) and (d) right columns are the original and the adaptively filtered interferograms, respectively.

V. Experimental Results

We investigate, in this section, the potential of the developed adaptive spatial multilooking technique with an emphasis on its application within MTInSAR processing chains (e.g., but not only [27], [39]) for the generation of ground deformation displacement maps. To this aim, we consider two SAR data sets collected at X- and L-band by the satellites of the Italian Space Agency COSMO-SkyMed (CSK) constellation and by the uninhabited aerial vehicle SAR (UAVSAR) system of the National Aeronautics and Space Agency (NASA)/Jet Propulsion Laboratory (JPL), respectively. For our experiments, in both cases, we use sets of SBs DInSAR multilook interferograms computed by using both the conventional boxcar multilooking filter and the proposed adaptive method.

The generated interferograms are then processed with the SBAS approach [27]. The change in average spatial coherence of the generated multilook interferograms due to the application of the innovative adaptive multilooking approach is then assessed.

A. Mt. Etna Case Study

We start our analysis by considering an SAR data set consisting of 146 SAR images of the Mt. Etna volcano area, Sicily, Italy, collected by the CSK satellites (ascending orbits) from July 2009 to January 2015. The distribution of the available SAR data in the temporal/perpendicular baseline plane, as well as the identified network of 453 SB data pairs used for the experiments, are shown in Fig. 7. For the selection of the SB data pairs a maximum perpendicular baseline of 800 [m] and the maximum time separation of 400 [days] have been considered.

Note that, for the effective use of EMCF [36] phase unwrapping procedure, a triangular shaped network of SB interferograms have been selected. To evaluate the potential of our multilooking method, we have not performed any prefiltering operation on the computed interferograms (e.g., by using the approach in [67]), even though the proposed method can be simply adapted to work with prefiltered (complex) interferograms. For the interferogram generation [34], we used precise satellite orbit information and a three-arc-second shuttle radar topography mission (SRTM) digital elevation model (DEM) of the region to remove the topographic phase contributions. First, we apply our adaptive spatial multilooking process to the computed single-look interferograms by producing multilooked DInSAR interferograms on the native full-scale single-look grid (see Section II).

To illustrate the effectiveness of the method, we consider two of the 453 SAR data pairs, 02-07-2011/03-08-2011 and 09-05-2012/10-06-2012, and compare the original single-look interferograms (Fig. 8(a) and (d), respectively) with multilooked interferograms obtained by using a conventional boxcar window [Fig. 8(b) and (e)] and our adaptive multilooking approach [Fig. 8(c) and (f)]. The comparison shows that our adaptive spatial multilooking technique significantly reduces the noise in the generated interferograms.

At finer scale, a comparison of the results is shown in Fig. 8(g)–(i), which shows the results for a small patch within the interferograms in Fig. 8(a)–(c). Specifically, Fig. 8(i) shows that the interferogram processed by our adaptive multilook algorithm both filters more noise and preserves more spatial details of the phase than the same interferogram generated using a conventional boxcar window [Fig. 8(h)]. To show the overall improved performance of the applied method, in Fig. 9 we compare the average spatial coherence for the stack obtained by applying adaptive multilooking to that obtained using the conventional boxcar window, and results show that the average spatial coherence obtained applying adaptive multilooking [Fig. 9(b)] exhibits a drastic increase

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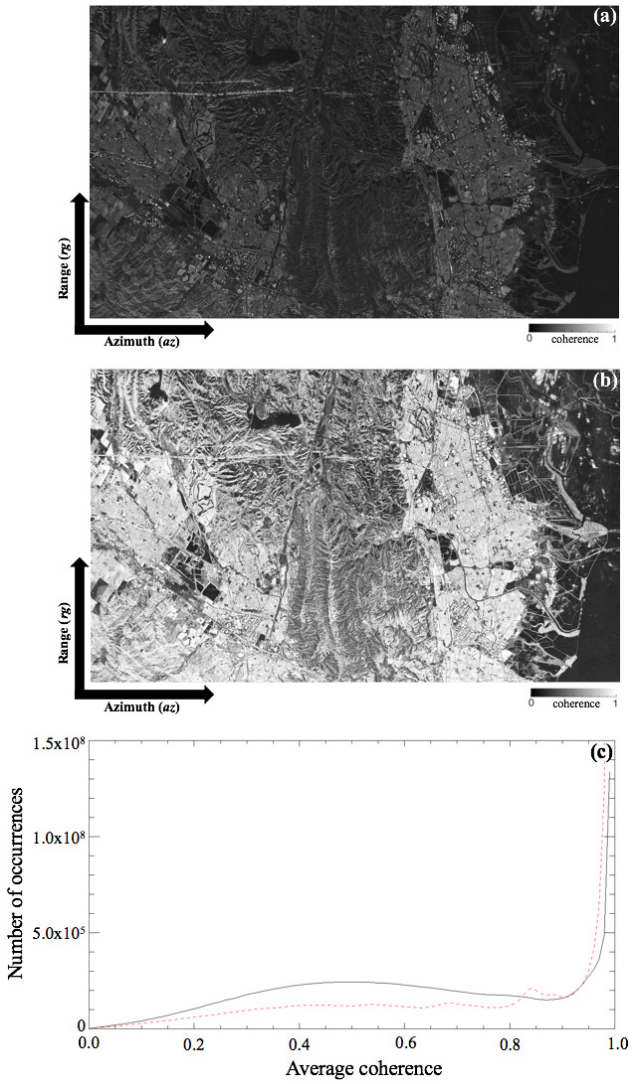


Fig. 13. Spatial coherence. (a) Original and (b) adaptively multilooked. (c) Histogram of original coherence (black continuous line) versus adaptively multilooked coherence (red dashed line) for the UAVSAR interferograms.

over boxcar averaging [Fig. 9(a)], with 42% of the pixels having higher coherence. The coherence improvement is particularly relevant at X-band, especially in the decorrelated, vegetated flanks of Mt. Etna, whereas the coherence of the old solidified lavas is present and also improved. Globally, the coherence increase is emphasized by the histograms of the average spatial coherence values shown in Fig. 10.

Finally, in order to evaluate a temporal analysis of deformation for Mt. Etna, we computed the mean deformation velocity map by processing the sequence of interferograms using the SBAS approach [27] before and after the application of the adaptive multilooking algorithm (see Fig. 11).

As a quality index of the phase retrieved ground displacement time-series, the temporal coherence factor is evaluated for every pixel. The latter is defined as [36]

$$\varsigma = \frac{1}{M}\left|\sum_{k=0}^{M-1} \exp[j(\phi_k - \check{\phi})_k]\right| \tag{22}$$

TABLE I
UAVSAR HAYWRD_23502 DATA. DATES ARE EXPRESSED IN THE FORMAT DAY/MONTH/YEAR

Acquisition n.	Date
0	20/02/2009
1	19/11/2009
2	15/01/2010
3	01/03/2010
4	23/04/2010
5	10/11/2010
6	12/07/2011
7	03/11/2011
8	13/04/2012
9	02/11/2012
10	07/05/2013
11	20/06/2014
12	12/11/2014
13	08/10/2015
14	24/03/2016
15	20/10/2017

where $\check{\phi}_k$ is the phase of the kth interferogram (at a given radar pixel) reconstructed from the computed displacement time-series. In our work, the well-processed SAR pixels are those for which the temporal coherence value is greater than 0.6.

The comparison of the two velocity maps shows that the one computed after applying the adaptive multilooking method [see Fig. 11(b)] is significantly more spatially dense than that computed without any multilooking process [see Fig. 11(a)].

B. Hayward Fault Case Study

We also applied the developed algorithm to a time series of L-band SAR images acquired by the UAVSAR instrument, an airborne system operated by the JPL for the U.S. NASA. The investigated area is east of San Francisco Bay, and includes parts of the towns of Fremont, California, to the west and Livermore, California, to the east. Table I contains a list of the dates of the 16 SAR data sets used. UAVSAR is an L-band instrument operating with 80-MHz bandwidth and deployed on a Gulfstream-3 aircraft [68]. The aircraft flight track is automatically controlled to keep the aircraft within a tube of 5 m radius centered on the planned track for $>= 90\%$ of the flight line. The aircraft flies at 12.5 km, the imaged swath is 22 km wide and the incidence angle varies from $\sim 22^\circ$ (near range) to 67° (far range). The UAVSAR flight track used in this analysis, Haywrd_23502, was oriented perpendicular to the Hayward fault.

We produced the stack of noise-filtered, multilooked interferograms for the period February 2009 to October 2017, and compared the results with and without the adaptive multilooking procedure applied. Fig. 12 shows two interferograms

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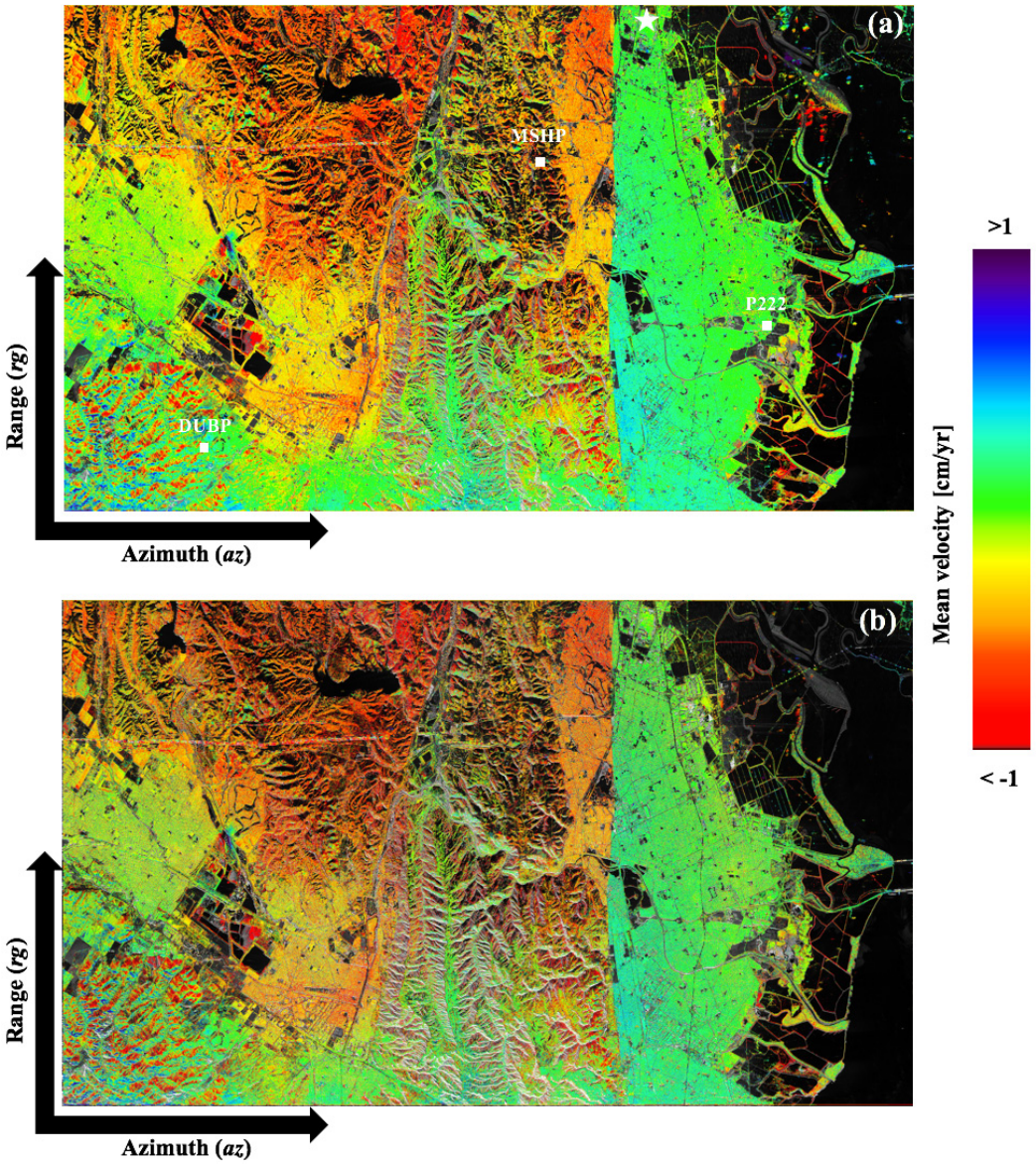


Fig. 14. Mean deformation velocity map of the east San Francisco Bay area showing the Hayward fault in the vicinity of Fremont, CA. (a) Before and (b) after applying the adaptive multilook algorithm to UAVSAR differential interferograms. The white star in (a) indicates the phase reference location.

computed before and after the application of the adaptive multilook operations. We also compare the average spatial coherence without [Fig. 13(a)] and with [Fig. 13(b)] the application of the proposed algorithm. Fig. 13(c) shows the histograms of the average spatial coherence before and after the application of the proposed method.

The SBAS algorithm was applied to the UAVSAR interferograms, and Fig. 14(a) and (b) shows the map of the mean deformation velocity of the Hayward fault area obtained using the original and the adaptively multilooked interferograms, respectively. From these results, we determined that over the set of high coherent points, the mean value of the deformation velocity difference between is 0.1 mm/year, while the standard deviation of the difference is 0.4 mm/year. A percentage increase of the well-processed SAR pixels of about 120% (from about 9 million to 11 million of points) was achieved. We would like to remark that, with respect to the X-band case, the UAVSAR surface deformation products obtained at L-band are generally more coherent, as expected, so the adaptive filtering has less impact on phase noise. Nonetheless, the increase of detectable pixels exhibiting very large coherence values is significant. Comparison of the X-band (Fig. 10) and L-band [Fig. 13(c)] histograms show that the L-band data showed a significant increase in pixels with coherence >0.97, which was not seen in the X-band data. The displacement time-series obtained by processing the noise-filtered interferograms was also compared with the measurements of the three GPS stations that are available in the study area, labeled to as DUBP, MSHT, and P222 in Fig. 14(a). Fig. 15(a) and (b) shows the plot of the relative displacement time-series between the DUBP and MSHT and DUBP and P222 stations, respectively, as seen by the GPS and SAR systems. The root mean square error (RMSE) of the difference between GPS and InSAR data is also reported in Fig. 15.

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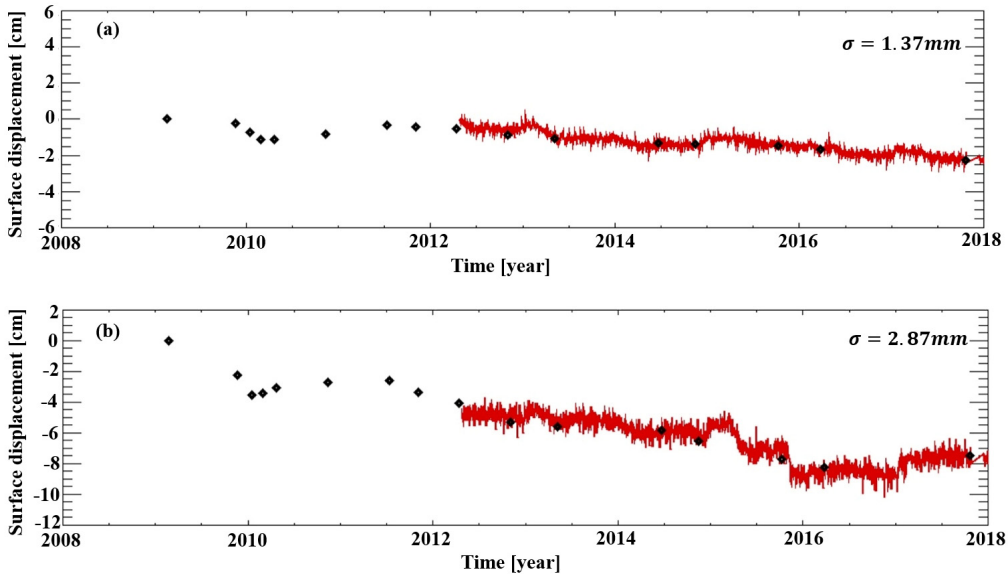


Fig. 15. Time-series for the InSAR-derived (red triangles) and GPS sensors' (black line) relative displacement between stations (a) DUBP and MSHT and (b) DUBP and P22. The standard deviation between the measurements are given for the period of overlap.

VI. Conclusion

In this work, we have proposed a solution for the generation of space–time adaptive multilooked differential SAR interferograms suitable for generation of ground deformation time-series using MTInSAR approaches. The main goal of the method is to reduce the decorrelation noise in medium-to-low coherent regions and thereby increase the number of usable SAR pixels. Rather than using the amplitude as a proxy for phase stability, a method is developed to evaluate statistical homogeneity based on the interferometric phase directly. The amplitude signals contain valuable, correlated information and the combined use of statistical tests relying on both amplitude and phase signatures is a possible avenue of future investigation. However, that work requires the development of the statistical framework that is comprehensively presented herein.

We would also like to remark that our research is not intended to demonstrate the superiority of the proposed method with respect to other already published approaches, but the aim is mainly to investigate the potential to use directly phase information and related statistics to identify SHPs and performs adaptive multilooking.

Results obtained with real data confirm the effectiveness of the approach. Experiments carried out using SAR data acquired at different operational wavelengths make it evident that the method developed increases the number of detectable SAR pixels, also in medium-to-high coherent regions. In the case of X-band, the achieved improvement in terms of detectable SAR pixels is about 42% and in the L-band case the improvement is about 120%. Globally, a significant increase in the average spatial coherence of the adaptive multilook interferograms is observed. The method can be complemented in the existing MTInSAR tools for the analysis of the ground deformation of DS targets, as demonstrated herein using SB methods although there is no restriction in using the proposed method in other MTInSAR frameworks. Extended investigations can assess the potential of the method at different microwave bands and for different geophysical processes that drive surface deformation.

Last, we note that the overall approach is computationally efficient, requiring independent analysis of neighboring patches within the SAR interferograms. For example, processing a sequence of 100 interferograms of dimension 1000×1000 using a processing architecture with an Intel Xeon CPU E5-2670 @ 2.60-GHz and 32-GB DDR3 (1600 MHz) takes about 30 min. The adopted scheme is suitable for easy implementation in a parallel computing environment, i.e., using high-performance computing (HPC) architectures. With that in mind, future activities will include the optimization of the different steps based on their efficient implementation using multicore parallelization.

Appendix A

A. Statistical Relationships Between the Coherence and the Circular Variance of a Directional Data Set

In this appendix, we summarize the statistical properties of the spatial coherence of an interferometric SAR data pair and the circular variance of the wrapped phase vectors. Our analysis starts from the basic observation that in interferometric processing, the conjugate multiplication of the two coregistered master and slave SAR images yields the complex interferogram

$$Z = S_1 \cdot S_2^* = |S_1| \cdot \left|S_2^*\right| \cdot \exp[j\phi] \tag{A.1}$$

where S_1 and S_2 are representative of the two complex-valued master and slave SAR images. To further reduce the phase noise corrupting the interferometric phase, a multilook operation [34] is usually carried out on the generated SAR interferograms, at the expense of degraded spatial resolution of the InSAR products. The multilooking operation consists in the calculation, pixel-by-pixel, of the statistical expectation value of the complex interferogram. Mathematically, under the hypothesis that SAR processes are stationary, jointly stationary

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and ergodic (in mean), the ensemble value can be substituted with spatial averages, thus resulting in the so-called sample covariance

$$\begin{aligned} Z_N &= E\left[S_1 \cdot S_2^*\right] = E[|S_1| \cdot \left|S_2^*\right| \cdot \exp[j\phi]] \\ &= \frac{1}{N}\sum_{Q\in\Omega} |S_1(Q)| \cdot \left|S_2^*(Q)\right| \cdot \exp[j\phi(Q)] \\ &= I_N \exp[j\bar{\phi}]. \end{aligned} \tag{A.2}$$

InSAR phase has been modeled as characterized by an additive noise [59], and the term ξ

$$\xi = \frac{E[S_1 \cdot S_2^*]}{\sqrt{E\left[|S_1|^2\right] \cdot E[|S_2|^2]}} = \gamma \exp[j\bar{\phi}] \tag{A.3}$$

represents the (complex) cross correlation between the two complex-valued SAR images forming the interferogram, whose amplitude γ expresses the so-called spatial coherence of the interferogram, and $\bar{\phi}$ is the corresponding multilook interferometric phase. If we consider the vector of wrapped phases $[\phi_0, \phi_1, \ldots, \phi_{N-1}]^T$ related to the N pixels into the multilooking window Ω, the statistical theory of directional data imposes that a measure of the phase dispersion of the vector with respect to its mean direction is given by the measure of the circular variance, which is defined as follows:

$$\upsilon = 1 - \rho = 1 - |E[\exp[j\phi]]| = 1 - \frac{1}{N}\left|\sum_{k=0}^{N-1} \exp[j\phi_k]\right|. \tag{A.4}$$

Here, we investigate the relationships between γ and υ. In [55], a noise model was derived for the (complex) interferometric phase

$$\exp[j\phi] = \cos(\phi) + j\sin(\phi). \tag{A.5}$$

Assuming the following additive phase model:

$$\phi = \bar{\phi} + \eta \tag{A.6}$$

where $\bar{\phi}$ is the original phase without noise, ϕ is the measured noisy phase and η is the additive noise. According to [50] and [55], it can be shown that

$$\rho = N_C = \frac{\pi}{4}\gamma\, F\left(\frac{1}{2}, \frac{1}{2}; 2, \gamma^2\right) \tag{A.7}$$

where N_C is the mean value of $\cos(\eta)$. Equation (A.7) shows that the two statistical quantities γ and ρ are similar, with γ being an upper bound for the mean resultant length ρ, thus it can be used as an interferometric phase quality parameter instead of the spatial coherence. Fig. 16 shows the comparison.

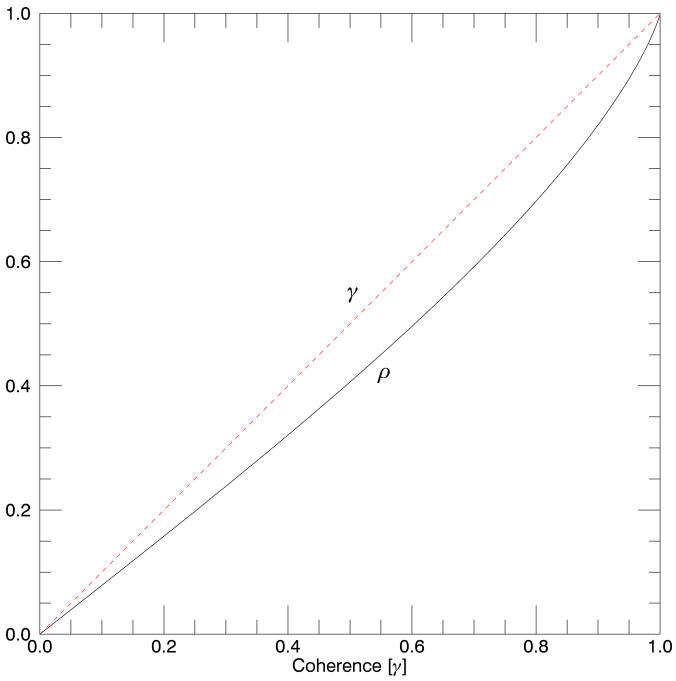


Fig. 16. Comparison between γ and ρ.

APPENDIX B

B. Considerations Regarding the Average Circular Coherence

In this appendix, we can demonstrate the validity of the following mathematical relationship:

$$\bar{\Gamma}_{\Omega'}(Q) \geq \bar{\Gamma}_W(Q) \quad \forall Q \in \Omega'. \tag{B.1}$$

To the purpose, we preliminarily observe that $\Omega' \subseteq \Omega$, where Ω and Ω' are the set of SHPs before and after the application of the Kuiper's test, respectively (see Section II). The average circular coherence calculated over the set Ω is

$$\begin{aligned} \bar{\Gamma}_\Omega &= \frac{1}{M}\sum_{k=0}^{M-1} \gamma_k^{ML} \\ &= \frac{1}{M}\sum_{k=0}^{M-1} \frac{1}{|\Omega|}\left|\sum_{Q\in\Omega} \exp\left[j\phi_k(Q)\right]\right|. \end{aligned} \tag{B.2}$$

The following relation also holds:

$$\begin{aligned} \Upsilon &= \frac{1}{M|\Omega|}\left|\sum_{Q\in\Omega}\sum_{k=0}^{M-1} \exp\left[j\phi_k^{\mathrm{HP}}(Q)\right]\right| \\ &= \frac{1}{M}\left|\sum_{k=0}^{M-1}\left[\frac{1}{|\Omega|}\sum_{Q\in\Omega} \exp\left[j\phi_k^{\mathrm{HP}}(Q)\right]\right]\right| \\ &= \frac{1}{M}\left|\sum_{k=0}^{M-1}\left[\gamma_k \exp\left[j\bar{\phi}_k^{\mathrm{HP}}(Q)\right]\right]\right| = \frac{1}{M}\sum_{k=0}^{M-1}\gamma_k \end{aligned} \tag{B.3}$$

where the last equality is valid because the average high-pass phase of the kth interferogram is zero by construction (see [50]), $\bar{\phi}_k^{\mathrm{HP}}(Q) = 0, K = 0, 1, \ldots, M-1, \forall Q \in \Omega$, and because the coherence values γ_k, $k = 0, 1, \ldots, M-1$ are definite positive. Accordingly, by equating (B.2) and (B.3) and by taking into account (12), we have

$$\begin{aligned} \bar{\Gamma}_\Omega = \Upsilon &= \frac{1}{|\Omega|}\left|\sum_{Q\in\Omega}\left\{\frac{1}{M}\sum_{k=0}^{M-1} \exp\left[j\phi_k^{HP}(Q)\right]\right\}\right| \\ &= \frac{1}{|\Omega|}\left|\sum_{Q\in\Omega} \bar{R}(Q)\exp[j\bar{\mu}(Q)]\right| \\ &\leq \frac{1}{|\Omega|}\sum_{Q\in\Omega}\bar{R}(Q). \end{aligned} \tag{B.4}$$

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At this stage, it is evident that the average spatial coherence $\bar{\Gamma}_\Omega$ is upper bounded by

$$\Theta_\Omega = \frac{1}{|\Omega|}\sum_{Q\in\Omega}\bar{R}(Q). \tag{B.5}$$

However, in practice, $\bar{\mu}(Q) \cong 0$. Accordingly, $\bar{\Gamma}_\Omega \cong \Theta_\Omega$. Similarly, it can be shown that $\bar{\Gamma}_W \cong \Theta_W$. The algorithm described in Section II (step 2) is based on progressively discarding those pixels in the moving averaging window W with $v(Q) = 1 - \bar{R}(Q) \leq \alpha$, where $v(Q)$ is the measured circular variance of the high-pass phase vector at generic pixel Q of the window W. Accordingly, if we compute the quantity (B.5), but on the whole window W, being $\Omega \subset W$, we have

$$\begin{aligned}\Theta_W &= \frac{1}{|W|}\sum_{Q\in W}\bar{R}(Q)\\ &= \frac{1}{|W|}\sum_{Q\notin\Omega}\bar{R}(Q) + \frac{1}{|W|}\sum_{Q\in\Omega}\bar{R}(Q)\\ &= \frac{1}{|W|}\sum_{Q\notin\Omega}\bar{R}(Q) + \frac{|\Omega|}{|W|}\Theta_\Omega.\end{aligned} \tag{B.6}$$

Because we progressively discard those pixels with the smallest values of $\bar{R}$, which is a definitive positive quantity, it is straightforward to prove that

$$\sum_{Q\notin\Omega}\bar{R}(Q) \leq (|W| - |\Omega|)\cdot\frac{1}{|W|}\sum_{Q\in\Omega}\bar{R}(Q). \tag{B.7}$$

By taking into account (B.6) and (B.7), the following relation holds:

$$\Theta_W \geq \left[\frac{|W| - |\Omega|}{|W|^2}\Theta_W + \frac{|\Omega|}{|W|}\Theta_\Omega\right] \tag{B.8}$$

and, after trivial mathematical manipulations, we have

$$\Theta_\Omega \geq \left[\frac{|W|}{|\Omega|} + \frac{1}{|W|} - \frac{1}{|\Omega|}\right]\Theta_W \geq \Theta_W. \tag{B.9}$$

Therefore, $\Theta_\Omega \geq \Theta_W$ and the more pixels are discarded, the larger is Θ_Ω. As a consequence, $\bar{\Gamma}_W \cong \Theta_W \leq \Theta_\Omega \leq \bar{\Gamma}_\Omega$. Note that after the application of the Kuiper's test, the cardinality of the SHPs (potentially) decreases. From [50], it can be shown that SAR pixels that pass the Kuiper's test have HP phases that are even more concentrated around their mean value. Accordingly, it is suitable to assume that $\bar{\Gamma}_W \cong \Theta_W \leq \Theta_\Omega \leq \Theta_{\Omega'} \cong \bar{\Gamma}_{\Omega'}$ Q.E.D.

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